

Qazi Saaheelur Rahaman

PHD STUDENT | COMPUTATIONAL BIOPHYSICS AND COMPLEX SYSTEMS

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Education

Aix-Marseille University (LAI, LIS, IBDM)

PHD – DEEP-LEARNING TENSORS OF CELL REARRANGEMENTS (DELETE)

Marseille, France

Oct. 2025 – Present

Indian Institute of Science Education and Research (IISER) Berhampur

BS-MS DUAL DEGREE

Berhampur, India

Dec. 2020 – Aug. 2025

Research Interests

Computational Biophysics | Cell and Tissue Dynamics | Active Matter | Image Analysis
Complex Systems | Adaptive Networks | Collective Dynamics | Machine Learning

Current Research

DEep-Learning TENSors of cell rearrangements (DELETE)

DOCTORAL RESEARCHER

Marseille, France

2025 – Present

- Developing quantitative and computational approaches to characterize cell and tissue rearrangements from imaging data.
- Combining image analysis, physical modelling and machine-learning tools to study collective dynamics and structural transitions in biological systems.
- Part of the SCHADOC MSCA Doctoral Programme

Publications

DARE: Division Axis and Region Estimation from 2D and 3D Time-Lapse Images

BIORxIV (SUBMITTED TO PLOS COMPUTATIONAL BIOLOGY)

2026

- Romain Karpinski, Alice Gros, Marc Karnat, **Qazi Saaheelur Rahaman**, Jules Vanaret, Mehdi Saadaoui, Sham Tlili, Jean-François Rupprecht. Equal contribution by the first four authors.
- [Link to the paper](#)

Noninvasive rheological inference from stable flows in confined tissues

ARXIV (SUBMITTED TO PRXLIFE)

2025

- Marc Karnat, Gautham Hari Narayana, Sudheer Kumar Peneti, Victoria Guglielmotti, **Qazi Saaheelur Rahaman**, Shreyansh Jain, Benoît Ladoux, Shao-Zhen Lin, Sham Tlili, René-Marc Mège, Jean-François Rupprecht
- [Link to the paper](#)

Collective dynamics of multiplex networks with distinct adaptation rules

PHYSICAL REVIEW E

2025

- Akash Yadav, **Qazi Saaheelur Rahaman**, V. K. Chandrasekar, and D. V. Senthilkumar.
- [Link to the paper](#)

Heterogeneous nucleation in a multiplex adaptive network

PHYSICAL REVIEW E

2025

- Akash Yadav, **Qazi Saaheelur Rahaman**, V. K. Chandrasekar, and D. V. Senthilkumar. Equal contribution by the first two authors.
- [Link to the paper](#)

Research Experience

Centre de Physique Théorique, Aix-Marseille University

RESEARCH INTERN

Marseille, France

May 2025 – July 2025

- Worked with the Out-of-Equilibrium Mechanics team on quantitative image analysis of cells and tissues.
- Supervised by Dr. Jean-François Rupprecht, CNRS Researcher.

Indian Institute of Science Education and Research (IISER) Thiruvananthapuram

MASTER'S THESIS

- Studied synchronization transitions and collective dynamics in multiplex adaptive networks, including heterogeneous nucleation and distinct adaptation rules.
- Research led to two publications in *Physical Review E*; supervised by Dr. D. V. Senthilkumar.

Thiruvananthapuram, India

June 2024 – May 2025

Institute of Cosmology and Gravitation, University of Portsmouth

RESEARCH INTERN

- Applied machine-learning regression methods to predict dark-matter halo properties from galaxy photometric and simulation-derived features.
- Developed scientific data-analysis workflows in Python under the supervision of Dr. Christopher Lovell.

Remote

Feb. 2023 – Sept. 2023

Scientific Activities

Oct. 2026

Short Talk

CENTURI Scientific Meeting

Jun. 2026

Poster

Soft Matter, Complex Fluids, Interfaces 2026

Jun. 2026

Participant

CENTURI Hackathon for Quantitative Biology

Apr. 2026

Participant

SCHADOC Doctoral School Spring School

Mar. 2026

Poster

EMBO Conference: AI and Biology

Oct. 2025

Workshop

Physical Biology of the Cell workshop by Rob Phillips

Cargese, Corsica

ICISE, Vietnam

CENTURI, Marseille

Carry-le-Rouet

EMBL, Heidelberg

CENTURI, Marseille

Awards and Scientific Outreach

2022

7th place, Round 1

International Natural Science Tournament

2021

Gold Medal

iGEM 2021; team nominated for Best Diagnostic Project

2021

Invited Student Talk

Turing Patterns – Mat-Sci Student Talk Series, Project Spiral

International

International

IISER Berhampur

Skills

Scientific Programming

Computational Methods

Research Tools

Design / Media

Python (NumPy, SciPy, pandas, Matplotlib, scikit-learn), C, Java, Matlab

Image analysis, machine learning, numerical modelling, network analysis

Jupyter, LaTeX, Git/GitHub, scientific visualization

Adobe Illustrator, Photoshop, Premiere Pro, After Effects